

Domain-Specific Knowledge Graphs in RAG-Enhanced Healthcare LLMs

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Abstract

Large Language Models (LLMs) generate fluent answers but can struggle with trustworthy, domain-specific reasoning. We evaluate whether domain knowledge graphs (KGs) improve Retrieval-Augmented Generation (RAG) for healthcare by constructing three PubMed-derived graphs: G1 (T2DM), G2 (Alzheimer’s disease), and G3 (AD+T2DM). We design two probes: Probe 1 targets merged AD–T2DM knowledge, while Probe 2 targets the intersection of G1 and G2. Seven instruction-tuned LLMs are tested across retrieval sources {No-RAG, G1, G2, G1+G2, G3, G1+G2+G3} and three decoding temperatures. Results show that scope alignment between probe and KG is decisive: precise, scope-matched retrieval, notably G2, yields the most consistent gains, whereas indiscriminate graph unions often introduce distractors that reduce accuracy. Larger models frequently match or exceed KG-RAG with a No-RAG baseline on Probe 1, indicating strong parametric priors, whereas smaller and mid-sized models benefit more from well-scoped retrieval. Temperature plays a secondary role; higher values rarely help. We conclude that precision-first, scope-matched KG-RAG is preferable to breadth-first unions, and we outline practical guidelines for graph selection, model sizing, and retrieval/reranking. Code and data available here: <https://github.com/sydneyanuyah/RAGComparison>. This paper was published in the 2025 IEEE International Conference on Big Data (BigData).